

IBR Model Quality Test Report

180 MW PV Plant | IEEE 2800.2 Conformity | 2026-06-26 05:34

Summary - Total 6 cases, 5 Pass / 1 Fail

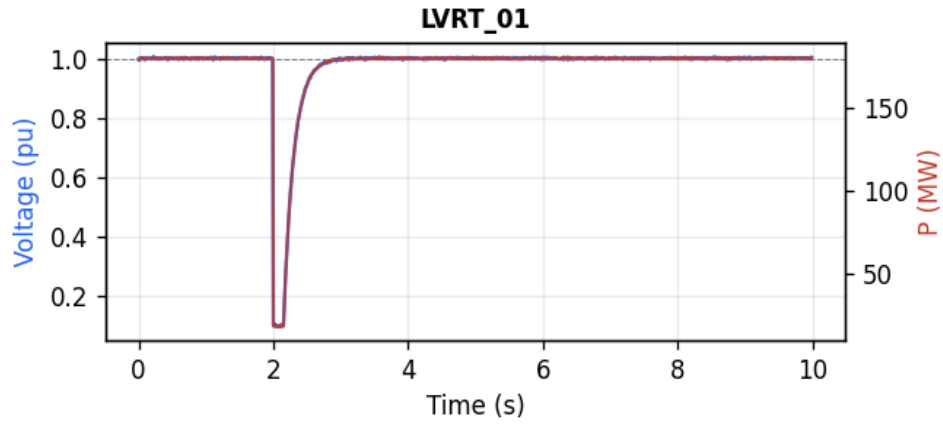
Test	Case	Verdict	P recovery	Max V (pu)
LVRT	LVRT_01	Pass	100%	1.006
LVRT	LVRT_02	Fail	52%	1.008
HVRT	HVRT_01	Pass	100%	1.154
HVRT	HVRT_02	Pass	100%	1.183
VSTEP	VSTEP_01	Pass	101%	1.056
VSTEP	VSTEP_02	Pass	99%	1.007

Low Voltage Ride-Through Test - LVRT_01

V=0.10pu, 0.15s @POI (IEEE2800 5.3.2)

Verdict: **Pass** (*rule-based*)

Reason: ■■ ■■ ■ ■■ ■■ ■■

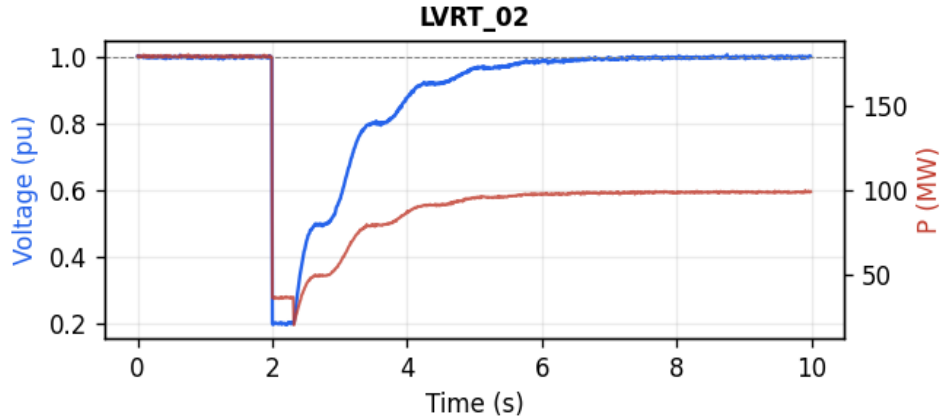


Low Voltage Ride-Through Test - LVRT_02

V=0.20pu, 0.32s SLG fault - ■■■ ■■■/■■■ ■■■ (Fail ■■■)

Verdict: Fail (rule-based)

Reason: ■■■■■ ■■■■ 52% < 95% (■■■ ■■■■); ■■■ ■■■ ■■■ ■■■■ (std 0.082pu, ■■■ ■■■)



Fail Reasons and Parameter Recommendations

1. Parameters to Modify (REGCAU1 & REECAU1 Model)

- **Tp** : Voltage filter time constant (voltage measurement filter)
- **Tiq** : Current filter time constant (or related delay)
- **Kvp** : Voltage proportional gain (local V control loop)
- **Kvi** : Voltage integral gain (local V control loop)

2. Current vs Recommended Values

Parameter	Model	Current	Recommended
Tp	REGCAU1	0.5	0.02
Tiq	REGCAU1	10.0	10.0
Kvp	REECAU1	0.1	0.9
Kvi	REECAU1	0.1	0.4

3. Why These Changes Are Required (WECC Generic Model Validation & Stability Improvement)

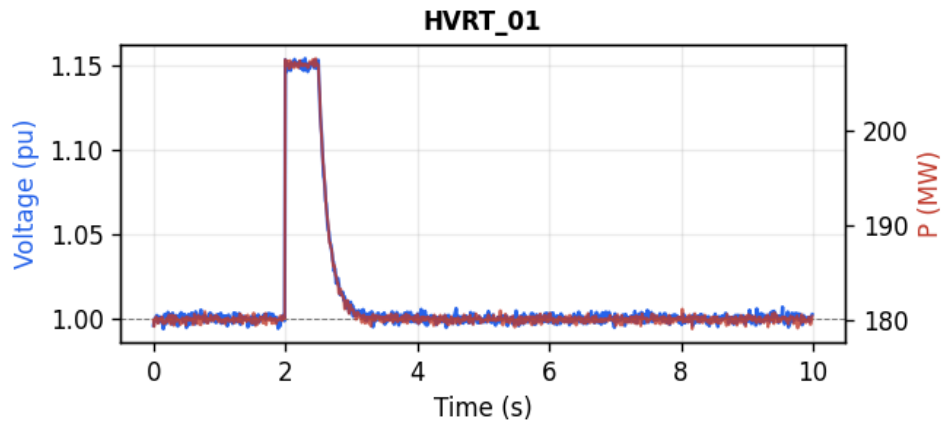
- Current values cause excessively slow / mismatched filtering and low gains, leading to sluggish response or insufficient damping in voltage/reactive power control during transients.
- High Tp (0.5 s) introduces excessive delay in voltage measurement, slowing down the control loop response.
- Raising Kvp/Kvi accelerates the local voltage control loop so active power recovers to $\geq 95\%$ of pre-fault output and post-fault voltage settles near 1.0 pu (WECC Generic Model Validation & IEEE 2800.2).

High Voltage Ride-Through Test - HVRT_01

V=1.15pu, 0.50s swell (IEEE2800 5.3.5)

Verdict: **Pass** (*rule-based*)

Reason: ■■ ■■ ■ ■■ ■■ ■■

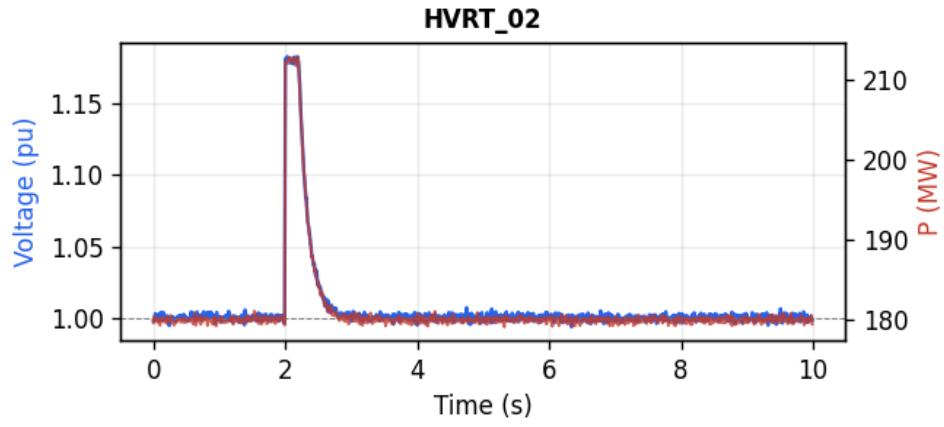


High Voltage Ride-Through Test - HVRT_02

V=1.18pu, 0.20s swell

Verdict: **Pass** (*rule-based*)

Reason: ■■ ■■ ■ ■■ ■■ ■■

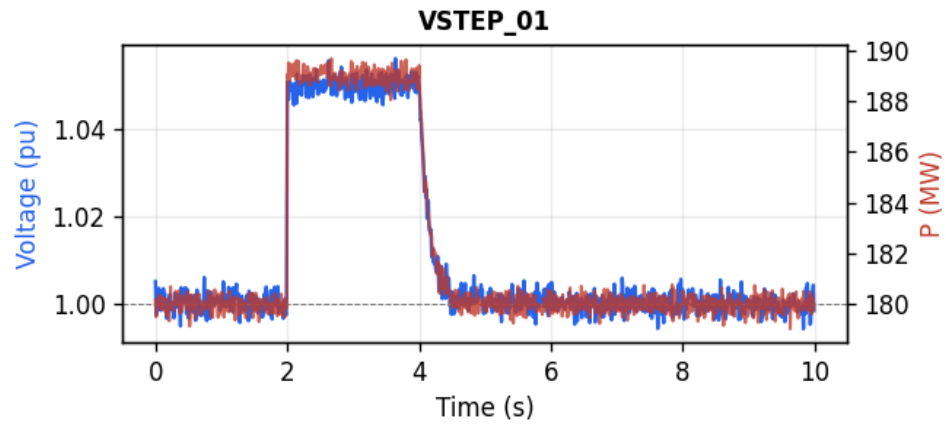


Voltage Step Change Test - VSTEP_01

+5% voltage step change response

Verdict: **Pass** (*rule-based*)

Reason: ■■ ■■ ■■■ ■■ ■■



Voltage Step Change Test - VSTEP_02

-5% voltage step change response

Verdict: **Pass** (*rule-based*)

Reason: ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

